Factor Timing Power of Flow Sentiment Signals

Abstract

The rise of exchange-traded funds (ETFs) has made thematic and sector investing low-cost and widely accessible. Based on the sector concentration of daily fund flows into non-broad-based equity ETFs, our prior study constructed the Flow Sentiment Index (FSI). In this paper, we extend the FSI application from market timing to factor timing by classifying style factors into aggressive categories (e.g., momentum and size) and defensive categories (e.g., low volatility and dividend), and constructing factor rotation rules based on month-end FSI signals. To ensure robustness, we conduct empirical tests from multiple dimensions. The empirical results suggest that the FSI-based rotation strategy modestly captures variation in market risk appetite and shows reasonable robustness across major broad-market universes. Composite factor construction helps reduce idiosyncratic noise and may improve signal stability. Overall, the FSI rotation approach offers modest improvements in portfolio risk-adjusted performance and provides a transparent and practical framework for dynamic style allocation in the A-share market.

1 Introduction

The rise of exchange-traded funds (ETFs) has made thematic and sector investing low-cost and broadly accessible. Our prior study indicates that fund flows in non-broad-based equity ETFs provide a unique window for observing real-time investor sentiment. By calculating the concentration of daily fund inflows into specific sectors (Herfindahl-Hirschman Index) and adjusting for fundamental factors such as market volatility and turnover, we construct a monthly Flow Sentiment Index (FSI). The core economic logic of the index is as follows: when investor sentiment is exuberant, capital concentrates into a handful of popular themes or sectors, driving up flow concentration (FSI > 0). Such sentiment-driven crowded trades are often

unsustainable and foreshadow declining future market returns. Conversely, when fund flows are dispersed (FSI < 0), it typically corresponds to subdued sentiment, and the market may possess upside potential. The empirical results are consistent with this hypothesis: FSI exhibits a significant negative predictive relationship with subsequent market returns, and this predictive power is stronger when ETF market share is high or when the growth style dominates. A simple long, flat market timing strategy based on FSI delivers substantial economic gains, with a Sharpe ratio that substantially surpasses the buy-and-hold benchmark. These findings suggest that FSI serves as a novel, high-frequency sentiment indicator grounded in trading behavior.

However, alpha in asset management stems not only from gauging the overall market direction, but also from precise exposure to specific styles or factors in different market environments. Factors such as value, momentum, quality, and low volatility have been shown to deliver long-term premiums, yet their performance exhibits pronounced cyclicality (Hotze et al., 2024). For instance, during bull markets or periods of rising risk appetite, aggressive factors like momentum and small size tend to perform strongly; whereas in bear markets or when risk aversion is elevated, defensive factors such as low volatility and high earnings quality become more favored. Investor sentiment is a core force driving the cyclical rotation of factors and exerts a significant cross-sectional impact on style returns such as size, momentum, growth, and low volatility (Baker & Wurgler, 2006).

Thus, having documented the FSI's predictive ability for market returns, a natural and highly practical question arises: Can the shifts in market sentiment and risk appetite captured by the FSI further guide more refined asset allocation—specifically, rotation among different style factors? If a high FSI level (exuberant sentiment) signals a buildup of overall market risk and a subsequent correction, then risk-sensitive, high-beta aggressive factors may face greater drawdown pressure, at which point defensive factors should be overweighted. Conversely, a low FSI level (subdued sentiment) may indicate a bottoming of market risk appetite and a subsequent recovery, at which point overweighting aggressive factors could generate higher

cyclical returns.

This study aims to extend the application of the FSI from market timing to factor timing, systematically examine whether fund flow sentiment can predict the cross-sectional returns of style factors, and construct an aggressive-versus-defensive factor rotation strategy based on FSI signals. We classify factors into aggressive (momentum, size, value) and defensive (low volatility, dividend, quality) categories, and establish a unified timing rule: if the month-end FSI is below zero, allocate to aggressive factors in the following month; if the FSI is greater than or equal to zero, allocate to defensive factors. To ensure robustness, we conduct empirical tests from multiple dimensions. The empirical findings provide evidence on the extent to which FSI-driven factor rotation can enhance portfolio performance across different market environments. Moreover, composite factor construction significantly reduces signal noise and enhances strategy stability. Overall, the FSI factor rotation strategy improves the portfolio’s risk-return profile and provides an effective and implementable research framework for the quantitative allocation of dynamic styles in the A-share market.

2 Research Design and Data Methodology

2.1 Sample and Data

The sample period in this paper is consistent with that of the original FSI study, spanning from June 2011 to June 2025, spanning from June 2011 to June 2025. The 14-year sample covers multiple market regimes in China's A-share market, including bull markets, sharp corrections, and periods of elevated volatility, facilitating a robust assessment of the strategy. All data are sourced from RiceQuant, a quantitative research platform in China. The research sample includes: monthly return data for all A-share stocks, the monthly Flow Sentiment Index (FSI) series, factor return data, and index constituent masks (CSI 500, CSI 800, and CSI 1000).

2.1.1 Flow Sentiment Index (FSI) Construction

The Flow Sentiment Index (FSI) is designed to capture the industry concentration of capital flows into non-broad-market equity ETFs, serving as a real-time proxy for investor sentiment. The construction follows a three-step procedure.

First, we estimate the daily flow for each ETF *j* on day τ as:

$$Flow_{j,\tau} = TNA_{j,\tau} - TNA_{j,\tau-1}(1 + R_{j,\tau}),$$

where $TNA_{j,\tau}$ is the total net asset value and $R_{j,\tau}$ is the return of ETF *j* on day τ.

Second, we compute the raw daily industry concentration of these flows using the Herfindahl-Hirschman Index (HHI):

$$HHI_\tau = \sum_{k=1}^{K} FlowShar\, e_{k,\tau}^2,$$

where $FlowShare_{k,\tau} = Flow_{k,\tau}/AggFlow_\tau$ is the share of industry *k* in aggregate ETF flows, with $AggFlow_\tau = \sum_{j=1}^{N_\tau} | Flo\, w_{j,\tau} |$. Industry-level flow $Flow_{k,\tau}$ is obtained by aggregating ETF flows using their portfolio weights in each industry.

Third, to isolate the sentiment-driven component from mechanical market conditions, we orthogonalize the raw concentration measure against contemporaneous market fundamentals. Specifically, we run a daily time-series regression of HHI_τ on a vector of market-level characteristics C_τ (the absolute market return, rolling 20-day market volatility, and daily change in market turnover) within a rolling window:

$$HHI_\tau = a + b^\top C_\tau + \epsilon_\tau.$$

The residual $\widehat{HHI_\tau}$ represents the neutralized concentration measure. Finally, the monthly FSI is constructed by averaging these daily orthogonalized residuals within each month:

$$FSI_t = \frac{1}{T_t}\sum_{\tau=1}^{T_t} \widehat{HHI_\tau},$$

where T_t is the number of trading days in month *t*. A positive FSI indicates that fund flows are more concentrated than market fundamentals would predict (exuberant sentiment), while a negative FSI signals abnormally dispersed flows (subdued sentiment).

2.1.2 Style Factors Classification

Style factors are classified into two categories according to standard conventions in academia and the investment industry: aggressive factors include value, size, and momentum; defensive factors include low volatility, dividend, and quality (Asness et al., 2013; Wurgler et al., 2010; Fama & French, 2014). The factor return data comprise long-short hedge portfolios and long-only portfolios, catering to different application scenarios.

This classification is grounded in the factors' distinct sensitivities to shifts in market risk appetite and macroeconomic conditions. Aggressive factors tend to perform strongly during periods of economic recovery, market expansion, and elevated risk appetite. Their excess returns primarily stem from compensation for bearing systematic risk, the correction of overly pessimistic market pricing, and the reversal of fundamental undervaluation (Fama & French, 2014). The value factor, constructed by going long high book-to-market stocks and short low book-to-market stocks (Fama & French, 2014), captures the premium associated with out-of-favor firms whose valuations rebound as market sentiment recovers. The size factor, measured by market capitalization and formalized in Fama and French's SMB factor, reflects the higher sensitivity of small-cap firms to improvements in economic growth expectations and liquidity conditions (Fama & French, 1988; Gertler & Gilchrist, 1994). The momentum factor, constructed by buying past winners and selling past losers over a formation period of approximately 12 months (Jegadeesh & Titman, 1993), benefits from trend continuation in rising markets but is vulnerable to severe crashes during market transitions. Value and momentum represent the most pervasive return premia across asset classes globally (Asness et al., 2013).

In contrast, defensive factors exhibit resilience during economic downturns, market declines, and periods of elevated volatility or contracting risk appetite. Their excess returns are attributed to stable operating fundamentals, sustainable cash flow streams, and the mispricing of low-risk assets under market irrationality (Ang et al., 2006; Blitz & Van Vliet, 2007). The low volatility factor, which goes long stocks with the lowest historical return volatility and short those with the highest, contradicts the traditional risk-return tradeoff and delivers superior risk-adjusted returns, particularly during market stress (Wurgler et al., 2010). The dividend factor is built by sorting stocks by trailing 12-month dividend yield, forming long-short portfolios of high-dividend versus low/zero-dividend stocks. High-dividend firms have mature, predictable cash flows and bond-like downside protection when risk appetite contracts, with minimal sensitivity to cyclical economic fluctuations (Fama & French, 1988). This stable, counter-cyclical profile defines its core defensive attribute. The quality factor, formalized as the Robust Minus Weak (RMW) portfolio in Fama and French (2014), captures the premium of firms with sustainable profitability, strong balance sheets, and low financial leverage. High-quality firms maintain stable earnings through downturns, and benefit from the market-wide flight-to-quality effect when aggregate risk appetite shrinks (Asness et al., 2018), placing it firmly in the defensive category.

This asymmetric return profile with respect to market sentiment provides the economic rationale for a rotation strategy between the two factor categories, where the FSI serves as the conditioning variable that signals the prevailing risk appetite regime (Baker & Wurgler, 2006).

2.2 FSI Factor Timing Strategy

This paper adopts a specific strategy logic: at the end of each month, the current FSI value is calculated. If FSI < 0, it indicates that market sentiment is within a reasonable range, fund flows are not extremely concentrated, and risk appetite is trending upward; this is identified as an aggressive state, and aggressive factors are

allocated in the following month. If FSI ≥ 0, it indicates that market sentiment is becoming overheated, capital is highly concentrated in a few themes, and risk appetite is about to decline; this is identified as a defensive state, and defensive factors are allocated in the following month.

2.3 Portfolio Construction and Testing Dimensions

This paper comprehensively tests the robustness of the strategy from four dimensions to ensure that the conclusions are not dependent on specific settings.

First, long-short portfolios versus long-only portfolios. The long-short portfolio captures the pure factor premium by going long the long leg and short the short leg, stripping out overall market fluctuations, and is suitable for academic testing. The long-only portfolio selects securities only within the long universe, aligning with the practical constraints of products such as public funds and separately managed accounts, thus possessing practical significance.

Second, equal weighting versus risk-parity weighting. Equal weighting represents the simplest portfolio construction method (DeMiguel et al., 2009), while risk parity represents a mainstream risk-controlled allocation approach. Comparing the two allows examination of the strategy's performance under different portfolio objectives.

Third, the full market versus broad-based index constituents. This paper repeats the strategy separately within the full A-share market, CSI 500, CSI 800, and CSI 1000 constituents. CSI 500 represents mid-cap stocks, CSI 800 represents large-cap plus mid-cap stocks, and CSI 1000 represents small-cap stocks, testing the strategy's applicability across different market-cap structures.

Fourth, single factors versus composite factors. To reduce the noise of individual financial metrics, this paper constructs composite factors using multiple indicators: a value composite factor (BP, EP, EV/EBITDA), a profitability composite factor (ROE, ROA, ROIC), and a growth composite factor (net profit growth, operating profit growth).

2.4 Performance Evaluation Metrics

To evaluate the strategy's performance comprehensively and scientifically, this paper adopts classical and widely accepted performance indicators common in academia and quantitative investment: annualized return, annualized volatility, Sharpe ratio, maximum drawdown, monthly win rate, and Calmar ratio. These metrics compare the performance differences between the FSI timing strategy and the benchmark portfolio from multiple dimensions, including return level, volatility risk, risk-adjusted return, extreme drawdown, and profit stability.

3 Empirical Results

3.1 Dimension 1: Basic Factor Timing Effectiveness — Long-Short vs. Long-Only Comparison

This section, based on the same set of FSI timing signals, tests the effectiveness of the basic factor timing strategy within a unified backtesting framework. It systematically compares the risk-return characteristics of the FSI long-short portfolio and the FSI long-only portfolio, with a non-timing multi-factor equal-weighted portfolio serving as the benchmark reference. The resulting performance metrics for the three portfolios are presented in Table 4-1.

Strategy	Annual Return	Volatility	Sharpe	Max Drawdown	Win Rate	Calmar Ratio	Total Return	t-stat	p-value
Long-Short	5.09%	9.61%	0.321	-22.97%	53.99%	0.221	96.20%	1.314	0.191
Long-Only	4.28%	19.33%	0.118	-36.16%	52.15%	0.118	76.68%	0.769	0.443
No-Timing Benchmark	4.72%	7.51%	0.362	-17.74%	59.51%	0.266	87.14%	1.424	0.156

Table 4-1 Strategy Performance Comparison

The performance results show the following: In terms of absolute return, the long-short portfolio delivers the highest return, while the long-only portfolio underperforms the static benchmark. In terms of risk, the long-only portfolio exhibits significantly higher volatility and drawdown risk, with volatility 2.6 times that of the benchmark and 2.0 times that of the long-short portfolio. In terms of risk-adjusted

return, the non-timing static benchmark performs best, followed by the FSI long-short portfolio. This indicates that, within the current sample period, the simple static equal-weighted multi-factor allocation has already achieved excellent risk-adjusted performance; the FSI rotation strategy enhances absolute return but has yet to surpass the benchmark in terms of the Sharpe ratio.

Overall, under the basic settings, the FSI signal demonstrates the ability to identify style regimes, and the long-short structure outperforms the long-only portfolio in risk-adjusted return. However, it should be noted that, in this sample period, the non-timing static equal-weighted multi-factor portfolio achieves the highest Sharpe ratio (0.362), while the Sharpe ratio of the FSI long-short rotation strategy is 0.321, not significantly surpassing the benchmark.

The t-statistics and corresponding p-values reported in Table 4-1 test the null hypothesis that the mean monthly return of each strategy is zero. For the Long-Short strategy, the t-statistic is 1.314 with a p-value of 0.191; for the Long-Only strategy, the t-statistic is 0.769 with a p-value of 0.443; and for the No-Timing Benchmark, the t-statistic is 1.424 with a p-value of 0.156. None of these p-values are below the conventional 0.05 threshold, indicating that, while the mean returns are positive, they are not statistically significantly different from zero at the 5% level. This suggests that, although the FSI signal can distinguish market states, under the current factor selection and simple rotation rules it has not yet been able to consistently outperform the static allocation in terms of risk-adjusted return. Nevertheless, the FSI long-short portfolio holds certain advantages in absolute return (5.09% vs. 4.72%) and drawdown control (-22.97% vs. -17.74%), indicating that optimized rotation rules still possess potential.

The net value and drawdown trajectories of the three portfolios are presented in Figures 4-1 and 4-2, respectively.

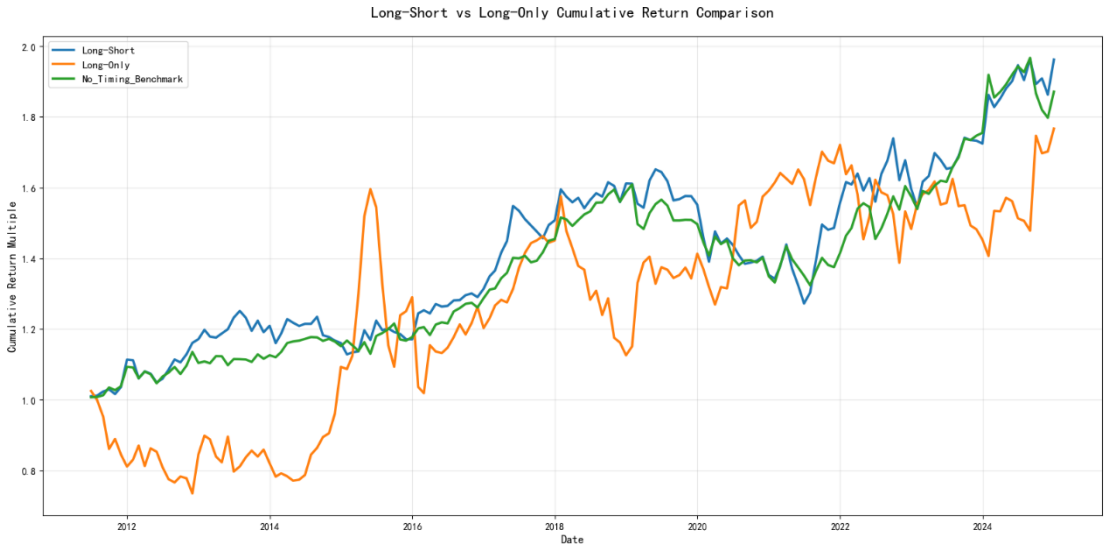


Figure 4-1 Long-Shrot vs Long-Only Cumulative Return Comparison

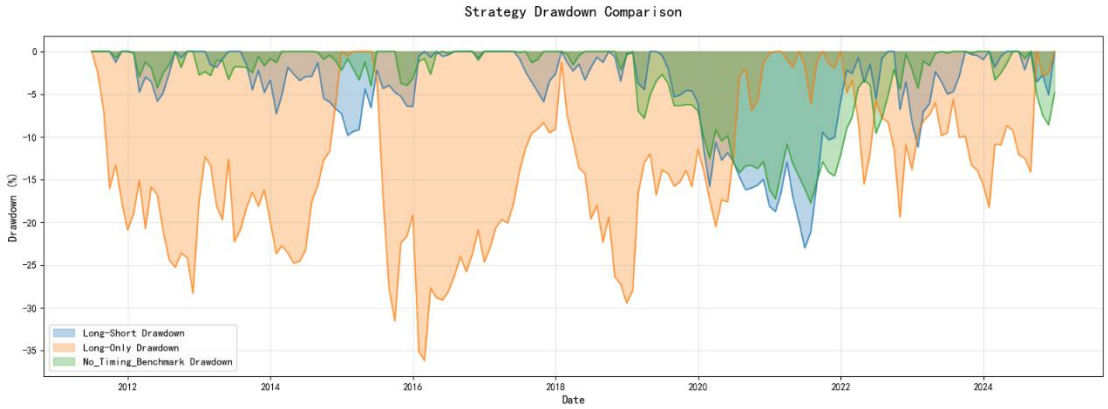


Figure 4-2 Strategy Drawdown Comparison

The cumulative net value curve shows that the FSI long-short portfolio follows the most stable trajectory, with long-term return performance materially higher than that of the long-only portfolio. The long-only portfolio experienced deep drawdowns during periods of severe market downturns, such as in 2015, 2018, and 2022, and its net value recovery efficiency was relatively low. The non-timing benchmark exhibited the smallest net value fluctuations, but its upside elasticity was insufficient, resulting in long-term returns that fell short of those of the long-short portfolio.

The dynamic drawdown analysis provides further evidence: the depth and duration of drawdowns for the long-only portfolio are significantly greater than those of the other two portfolios. Through its hedging mechanism and defensive factor

switching, the long-short portfolio demonstrates strong risk control capability during market downturns (Ang et al., 2006), with both drawdown magnitude and recovery speed superior to those of the long-only portfolio, lagging behind the non-timing benchmark only in the ultimate stability of drawdown control.

Taken together, the results in this section suggest that the FSI signal carries information about market states and can enhance the portfolio's downside resilience through aggressive-defensive factor rotation. The long-short hedging structure appears to be a more effective vehicle for factor timing strategies. Compared with the long-only form, the long-short portfolio performs better across dimensions such as return, volatility, maximum drawdown, and risk-adjusted return.

3.2 Dimension 2: Comparative Analysis of Equal Weighting and Risk-Parity Weighting

In this section, under a unified backtesting framework and FSI timing signals, we conduct robustness tests on the factor portfolio weighting methodology. The focus is on comparing the performance differences between the equal-weighting and risk-parity allocation methods, with the non-timing multi-factor equal-weighted portfolio serving as the benchmark, in order to examine the impact of the weighting rule on strategy effectiveness. It should be noted that, due to data limitations, the originally planned true market-cap weighting at the stock level cannot be implemented; it is therefore replaced by a risk-parity approach based on monthly factor return series, weighted by the inverse of the rolling 24-month volatility, with the weighting adjustment completed solely at the factor rotation level. The resulting performance metrics for the three portfolios are presented in Table 4-2.

Strategy	Annual Return	Volatility	Sharpe	Max Drawdown	Win Rate	Calmar Ratio	Total Return	t-stat	p-value
Equal Weight	5.09%	9.61%	0.321	-22.97%	53.99%	0.221	96.20%	1.314	0.191
Risk Parity	3.54%	9.60%	0.160	-19.78%	49.64%	0.179	49.59%	0.687	0.494
No-Timing Benchmark	4.72%	7.51%	0.362	-17.74%	59.51%	0.266	87.14%	1.424	0.156

Table 4-2 Portfolio Strategy Risk-Return Performance Comparison

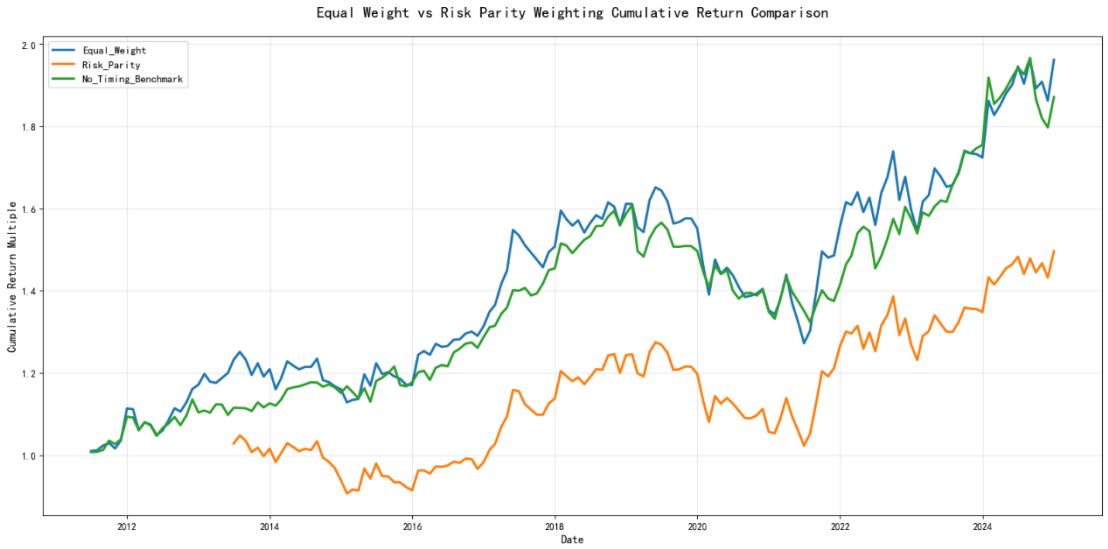


Figure 4.3 Equal Weight vs Risk Parity Weighting Cumulative Return Comparison

It is clearly evident from the backtest results that the equal-weighted strategy outperforms the risk-parity strategy in terms of annualized return, Sharpe ratio, and win rate. This suggests that simple equal-weighted allocation is sufficient to effectively capture the returns from factor rotation, while the complex inverse-volatility weighting dilutes the portfolio's overall return. This result aligns with the classic conclusion of DeMiguel et al. (2009): in real markets characterized by estimation errors and parameter uncertainty, simple and parsimonious allocation rules often exhibit greater out-of-sample robustness than complex optimization methods. Although the risk-parity strategy contained the maximum drawdown at -19.78% (slightly better than the equal-weighted -22.97%), its annualized return declined substantially, resulting in a Sharpe ratio of merely 0.160. Formal statistical tests confirm that neither the equal-weighted strategy ($t = 1.314$, $p = 0.191$) nor the risk-parity strategy ($t = 0.687$, $p = 0.494$) achieves a significantly higher mean return than the static benchmark. Notably, both Sharpe ratios fall below that of the non-timing benchmark (0.362). This signals that, in the specific market environment of this test round, simple factor rotation did not deliver risk-adjusted returns that surpass static holding.

However, due to data limitations, this study could only implement the risk-parity weighting test at the factor rotation level and was unable to conduct a true market-cap

weighting strategy test at the factor construction level. To perform a true market-cap weighting strategy test, three types of core data would need to be acquired: stock-level returns, stock market capitalizations, and individual stock factor exposures, and the market-cap weighting rule would need to be applied at the factor construction stage rather than adjusting weights solely at the factor rotation stage. Furthermore, although in this test period the simple monthly factor rotation did not statistically outperform the static equal-weighted allocation in terms of mean returns, it exhibited advantages in absolute return (annualized return) and downside risk control (maximum drawdown)—the equal-weighted strategy achieved a higher return, while the risk-parity strategy recorded a smaller drawdown. This demonstrates that the FSI signal can differentiate between market states. Future optimization should focus on transforming this capability into refined factor allocation weights, for example, by incorporating volatility adjustments or employing dynamic weighting based on signal strength, rather than relying on simplistic all-in or all-out switches.

3.3 Dimension 3: Factor Timing Effectiveness across Different Broad-Based Stock Universes

To test the robustness of the strategy under different market-cap structures, this paper replicates the empirical analysis separately within the constituent stocks of the CSI 500, CSI 800, and CSI 1000 indices. These three indices represent mid-cap, large-plus-mid-cap, and small-cap stocks, respectively, covering the main market-cap segments of the A-share market.

Strategy	Volatility	Sharpe	Sortino	Max Drawdown	Calmar Ratio	Win Rate	Annual Return	t-stat	p-value
zz500 Timing	18.29%	0.547	0.749	-26.13%	0.459	30.77%	12.00%	2.251	0.026
zz500 Benchmark	25.12%	0.108	0.164	-53.11%	0.089	52.07%	4.70%	0.856	0.393
zz800 Timing	17.18%	0.560	0.777	-25.16%	0.462	30.77%	11.63%	2.283	0.024
zz800 Benchmark	23.67%	0.087	0.133	-49.95%	0.081	51.48%	4.06%	0.755	0.451
zz1000 Timing	22.12%	0.504	0.657	-23.50%	0.560	31.25%	13.15%	1.879	0.063
zz1000 Benchmark	29.02%	0.076	0.120	-59.22%	0.071	50.00%	4.19%	0.704	0.482

Table 4-3 Performance of Timing and Benchmark Strategies Across CSI 500, CSI 800 and CSI 1000

Table 4-3 reports the performance of the FSI timing strategy across the three

major broad-based indices. From a return perspective, the annualized returns of the FSI timing strategy for the CSI 500, CSI 800, and CSI 1000 reach 12.00%, 11.63%, and 13.15%, respectively, while the corresponding buy-and-hold benchmarks achieve annualized returns of only 4.70%, 4.06%, and 4.19%. The strategy's return is roughly two to three times that of the benchmark. The Newey-West adjusted t-tests indicate that the return differences are statistically significant at the 5% level for the CSI 500 ($t = 2.251$, $p = 0.026$) and CSI 800 ($t = 2.283$, $p = 0.024$) indices. For the CSI 1000 index, the return differential is marginally significant at the 10% level ($t = 1.879$, $p = 0.063$).

In terms of risk control, the timing strategy does not systematically reduce the maximum drawdown: the CSI 500 timing drawdown is -26.13%, compared to the benchmark's -53.11%; the CSI 800 timing drawdown is -25.16%, compared to -49.95%; and the CSI 1000 timing drawdown is -23.50%, compared to -59.22%. While the absolute drawdown levels of the timing strategy are substantially lower than those of the benchmarks across all three indices, formal statistical tests on drawdown severity are not conducted here, and the drawdown comparisons should be interpreted with caution. Overall, FSI timing delivers a substantial return enhancement across all three indices, with statistical significance confirmed in two of the three cases, while also showing meaningfully lower maximum drawdowns relative to the benchmarks.

To further examine the strategy's robustness across different market-cap structures, factor rotation backtests are conducted separately within the constituent universes of the CSI 500, CSI 800, and CSI 1000. The results show that FSI timing does not depend on a specific market-cap structure and remains stable and effective across all indices.

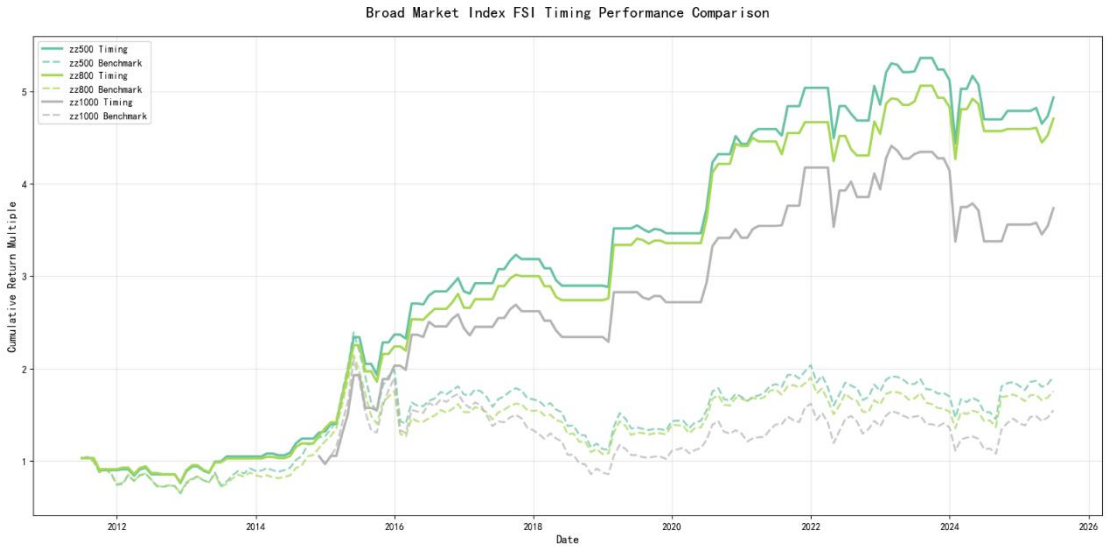


Figure 4-4 Broad Market Index FSI Timing Performance Comparison

In summary, the empirical results demonstrate that the FSI timing strategy substantially outperforms the buy-and-hold benchmark in terms of mean returns across all three broad-based indices, with the differences being statistically significant for the CSI 500 and CSI 800 at the 5% level. This superior return performance does not appear to be confined to a single market-cap segment, consistent with the interpretation that the FSI captures a systemic shift in market-wide risk appetite rather than noise from localized sectors. These findings provide preliminary support for the application of FSI-based factor timing across different market-cap segments.

3.4 Dimension 4: Construction and Effectiveness of Multivariate Composite Factors

Finally, to mitigate the noise and perturbations introduced by individual indicators, this paper further constructs three composite factors: a value composite factor, a profitability composite factor, and a growth composite factor. To verify the predictive power of the composite factors, we calculate the information coefficient (IC) between each factor and the subsequent period's return. The mean IC measures the direction and strength of the factor's predictive ability, the IC IR (information ratio) measures the robustness of the prediction, and the proportion of IC > 0 reflects

the factor's success rate in generating excess returns.

Factor	IC Mean	IC Std	IC IR	IC Positive Ratio	t-stat	p-value
Valuation Composite	0.054	0.152	0.354	63.50%	5.854	0.000
Profitability Composite	0.014	0.151	0.090	52.36%	1.483	0.139
Growth Composite	0.011	0.104	0.110	50.75%	1.794	0.074

Table 4-4 IC Performance of Valuation, Profitability and Growth Factor Composites

From the overall test results, all three composite factors exhibit positive mean ICs, indicating a directional ability to predict stock returns. The value composite factor demonstrates the strongest predictive performance, with a mean IC of 0.054 that is highly statistically significant ($t = 5.854$, $p < 0.001$). The profitability composite factor yields a mean IC of 0.014, which is not statistically significant at conventional levels ($t = 1.483$, $p = 0.139$). The growth composite factor registers a mean IC of 0.011, which is marginally significant at the 10% level ($t = 1.794$, $p = 0.074$). The multivariate composite construction approach achieves the smoothing of the factor IC series for all three factors, effectively mitigating the volatile impact of individual indicators.

The empirical results from the IC analysis indicate that the constructed composite factors, especially the value composite factor, exhibit a consistent positive predictive ability for future stock returns. This suggests that the multivariate synthesis approach is effective in reducing noise and enhancing signal stability, providing a useful foundation for future factor rotation research that could employ these self-constructed composite factors as the underlying investment universe.

4 Conclusions and Implications

4.1 Main Conclusions

Building on the Flow Sentiment Index (FSI) proposed in our prior research, this paper extends its application from market timing to factor timing, systematically investigates the predictive ability of the FSI for the rotation between aggressive and defensive factors, and constructs a unified monthly rotation strategy. Through multi-

dimensional empirical analysis, this paper reaches the following three core findings:

First, the FSI shows modest but measurable factor timing ability across portfolios and universes. The long-short structure tends to achieve better risk-adjusted performance than the long-only format, though formal statistical significance is mainly observed in broad-market index universes rather than in long-short or long-only standalone tests. Moreover, FSI-driven factor rotation achieves pronounced return enhancement and drawdown control effects across the CSI 500, CSI 800, and CSI 1000 broad-based indices. This demonstrates that the FSI strategy adapts well to mid-cap, large-plus-mid-cap, and small-cap broad-based indices, exhibiting strong universality.

Second, portfolio construction influences strategy outcomes, and the long-short hedge portfolio appears better suited to isolating factor timing benefits than the long-only portfolio. The research finds that, under the same FSI signals, a factor rotation strategy constructed with long-short hedge portfolios significantly outperforms the long-only version in core performance metrics such as the risk-return ratio (Sharpe ratio) and maximum drawdown control. This confirms the theoretical expectation: by stripping out market systematic risk through hedging, the long-short portfolio can capture the rotation patterns of factor premiums more purely and efficiently (Asness et al., 2013), making it a more effective tool for implementing the factor timing concept.

Third, constructing composite factors through multivariate synthesis can enhance factor quality, providing more robust underlying assets for the rotation strategy. By integrating multiple individual indicators, composite factors effectively reduce idiosyncratic noise and smooth factor return volatility, laying a useful foundation for further refining the factor rotation framework.

4.2 Practical Implications

The conclusions of this paper offer practical reference value. With its publicly

available data, simple rules, and high reproducibility, the FSI can be embedded as an independent sentiment-style module into existing multi-factor investment systems. Specifically, when the FSI signal is bullish, moderately overweight exposure to aggressive factors (e.g., momentum, small size); when the FSI signal is bearish, shift toward defensive characteristics such as low volatility and high dividends. This "factor rotation overlay" does not require changes to the underlying stock selection model and can be easily implemented in real investment systems. It enables dynamic adjustment of aggressive and defensive style factor exposures based on sentiment signals, offering a potential approach to return enhancement and drawdown control in mainstream broad-based index enhancements such as the CSI 500, CSI 800, and CSI 1000. Through long-only or long-short hedging structures, it can be adapted to the operations of public funds, hedge fund managed accounts, and flexible allocation products. Additionally, it can provide a basis for gauging market risk appetite in FOF macro asset allocation and portfolio risk management. The validated approaches to composite factor construction, equal-weighted allocation, and long-short portfolio design can also serve as practical references for the development of quantitative factor investment systems in the A-share market.

4.3 Research Limitations

Despite the multi-dimensional robustness checks conducted in this paper, certain limitations remain due to data constraints and research design. First, data constraints prevented the construction of true market-cap-weighted factors and the incorporation of transaction costs. Second, the simplified binary rotation rule treats FSI as a mere on/off switch, ignoring the continuous strength of sentiment signals and the potential for nuanced sub-factor weight optimization, which limits the strategy's flexibility. Third, the study is confined to the A-share market and a traditional set of factors, without cross-market validation or coverage of frontier factors, leaving the generalizability of the findings in question. Finally, the composite factors were only tested for their stock selection ability and were not integrated into the rotation framework, so their potential to further enhance the strategy remains unrealized.

4.4 Future Research Directions

Based on the above limitations, future research can be advanced in the following directions. First, the binary switch can be replaced by a gradual allocation model that maps continuous FSI values and historical percentiles to dynamic position weights. Incorporating volatility targeting and drawdown constraints would further smooth factor exposures, improving both the Sharpe ratio and the strategy's operational stability. Second, the value, profitability, and growth composite factors should be integrated into the FSI rotation system in place of single factors, and a granular transaction cost model that reflects the liquidity differences between large- and small-cap A-shares should be developed to bring backtest results closer to live trading conditions. Third, the framework can be extended to mature markets such as Hong Kong and U.S. equities for cross-market validation, while macro cycle indicators, option-implied volatility, and text-based sentiment measures can be combined to construct a more robust composite sentiment timing system, thereby enhancing the reliability and predictive accuracy of factor rotation.

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